16

Universal properties

The idea of characterizing something by the role it plays in context. Looking for extremities in a particular category, key to the whole idea of category theory.

16.1 Role vs character

Characterizing things by the role they play in a particular context is different from thinking about their intrinsic characteristics. We have seen a few examples of this already. We could describe 0 as a number representing nothing, which is describing it by an intrinsic characteristic. Or we could describe it as the unique number such that when you add it to any other number nothing happens. Not only does 0 play this role but, crucially, it is the *only* number that plays this role, so we can describe it by that role and all know we're talking about the same number.

In normal language we often use words that mix up whether we are talking about a role or an intrinsic characteristic. I mentioned some examples in Chapter 1, including pumpkin spice: I used to think it was spice made of pumpkins, when in fact it is a spice typically used in pumpkin pie. However, it is now used in all sorts of other things that don't involve pumpkin, including (infamously) pumpkin spice lattes.[†] A more pernicious example is "bikini body" which comes with the (largely sexist) idea that you need to be slim and toned (intrinsic characteristic) to wear a bikini, whereas, as some amusing memes put it, all you need to do to get a bikini body is get a bikini and put it on your body.

When those linguistic quirks are accidents of history rather than signs of misogyny I enjoy their quirks, but either way they make me think of category theory. I am intrigued by how deeply we have internalized some of this language so that we might not notice how contradictory it is deep down, but it can be confusing to people learning a new language, or moving to a different country.

[†] I hear that some brands of pumpkin spice latte do now have pumpkin as an ingredient. I'm sure if I don't include this footnote someone will write and tell me I made a mistake.

Things To Think About

T 16.1 Think about the following and how the words we use are mixing up role and intrinsic character, and in what way they do and don't make sense. Note that some are particular to American English.

1. baseball cap
2. film
3. hang up the phone
4. pound cake
5. biscuit (from the French "bis cuit")
6. fat-free half-and-half

It interests me that food items often have quirks of language in them especially if they are traditional things that have been passed down through generations and possibly they mutated as they went along.

Things To Think About

T 16.2 Now for some more mathematical examples. We have seen how to characterize the number 0 by a role rather than a property. What about the number 1? -1? i? Do these characterizations specify the number uniquely?

The number 1 is the unique number such that when we multiply it by any other number nothing happens. Formally we can say:[†]

$$\exists!\, a \in \mathbb{R} \quad \text{such that} \quad \forall x \in \mathbb{R} \;\; ax = x.$$

Every part of this definition is important and we must get all the clauses in the right order. If we just look at the end part "$ax = x$" we might be tempted to say this is true when $x = 0$. Now, "$ax = x$" is indeed true when $x = 0$, but we're not supposed to be looking for one value of x where this is true, we're supposed to be looking for one value of a that makes this true for *all* x. The only number that works is 1, so 1 is the unique number with this property.

This characterizes the number 1 as the *multiplicative identity*, and shows a way in which 1 is analogous to 0, as 0 is the *additive identity*. When we characterize things by universal properties one of the points is to find things that play analogous roles in different contexts.

We can characterize the number -1 as the additive inverse[‡] of 1, and that pins it down uniquely. However the imaginary number i is more subtle. It is defined as a square root of -1, that is, by it satisfying the equation $x^2 = -1$. However it is not the *unique* number with that property: $-i$ also satisfies the equation. This results in the curious fact that if we think about roles then we can't tell the difference between i and $-i$. This is a favorite weird conundrum

[†] Translation: there exists a unique a in the reals such that for all x in the reals $ax = x$.

[‡] That is: $1 + (-1) = 0$.

of mine in abstract mathematics. We have to arbitrarily pick one and call it i, and then the other one is $-i$. However, as we can't tell them apart until we've picked one, we also have no way to say which one we've picked. The whole thing should feel very disconcerting, maybe a bit quantum.

The question of how uniquely we can pin things down is an important question where universal properties are concerned. We may find several objects that all satisfy the property, but they will all be *uniquely isomorphic* so the category "can't tell them apart". Thinking categorically we should embrace not being able to tell them apart, even though there are several of them.

To me this is a lot like treating people equally even though they are different. It does lead to a linguistic conundrum where we're not sure whether to say "a" or "the" for something that is categorically unique but not literally unique. This can be frustrating to non-categorically minded people, or people who are not (yet) in the spirit of category theory. We are also liable to use the word "is" in a categorical way to say that something "is" something else if it is categorically so, rather than literally so.

Anyway, when we pin things down by a universal property, we look at the roles things play in context, rather than their intrinsic characteristics, and then we look at the extremities.

16.2 Extremities

Sometimes we look for things that play a certain role in context, and then we look for the one that is most extreme among all of those. For example, to find the *lowest common multiple* of two numbers we look at all the common multiples, and find the lowest one. This is what we do notionally anyway; in practice there are ways to calculate it.

Looking for extremities is one way to examine the inner workings of a situation and make comparisons. For example, if we look at the tallest mountain in the UK we find that it is really not very tall compared with the tallest mountain in the US. This gives us a rudimentary hint of the difference in general scale between the two countries.

However, we can also do it the other way round: we could start with an object we're trying to study, and then look for a context in which it is extreme. This then tells us something about the nature of that context. For example if I am ever the tallest person in the room then it shows it's a room full of rather short people, or children. Whereas if I'm the shortest person in the room it's not saying much; it might just be saying I'm in a room full of men, which happens quite often at math events.

In the cube diagrams from earlier, 30, 42 and rich white men occupy an analogous extreme position, in this case at the top because of how I've drawn the cubes.

$$\begin{tikzcd}[column sep=small]
 & \mathbf{30} \arrow[dl] \arrow[d] \arrow[dr] & \\
6 \arrow[d] \arrow[dr] & 10 \arrow[dl] \arrow[dr] & 15 \arrow[dl] \arrow[d] \\
2 \arrow[dr] & 3 \arrow[d] & 5 \arrow[dl] \\
 & 1 &
\end{tikzcd}
\qquad
\begin{tikzcd}[column sep=small]
 & \mathbf{42} \arrow[dl] \arrow[d] \arrow[dr] & \\
6 \arrow[d] \arrow[dr] & 14 \arrow[dl] \arrow[dr] & 21 \arrow[dl] \arrow[d] \\
2 \arrow[dr] & 3 \arrow[d] & 7 \arrow[dl] \\
 & 1 &
\end{tikzcd}
\qquad
\begin{tikzcd}[column sep=small]
 & \textbf{rich white men} \arrow[dl] \arrow[d] \arrow[dr] & \\
\text{rw} \arrow[d] \arrow[dr] & \text{rm} \arrow[dl] \arrow[dr] & \text{wm} \arrow[dl] \arrow[d] \\
\text{r} \arrow[dr] & \text{w} \arrow[d] & \text{m} \arrow[dl] \\
 & \varnothing &
\end{tikzcd}$$

It is more rigorous to identify the extremity relative to the arrows in the category, rather than relying on the physical positioning on the page. We can say that 30, 42 and rich white men are each in a position where all arrows begin (with the direction for arrows that I've chosen here). By contrast we could look at the place where all arrows end: 1, and non-rich non-white non-men.

Things To Think About

T 16.3 What can we say about extremes in each of the following situations? Is there an extreme where arrows begin? Is there an extreme where arrows end?

1. The natural numbers expressed like this: $0 \longrightarrow 1 \longrightarrow 2 \longrightarrow 3 \longrightarrow 4 \longrightarrow \cdots$
2. The integers like this: $\cdots \longrightarrow -2 \longrightarrow -1 \longrightarrow 0 \longrightarrow 1 \longrightarrow 2 \longrightarrow \cdots$
3. A category with two objects and no arrows: $\begin{tikzcd} \bullet & \bullet \end{tikzcd}$
4. A category that goes round in circles: $\begin{tikzcd} \bullet \arrow[r, bend left] & \bullet \arrow[l, bend left] \end{tikzcd}$

The natural numbers expressed in a line has all its arrows starting from 0. However, it has no ending place for the arrows as the natural numbers keep going "forever" — there is no biggest natural number. The integers keep going forever in both directions so they have no starting point or ending point.

In the example with no arrows the objects are completely unrelated so neither of them seems like an extreme. The category in a circle also doesn't seem to have an extreme but for the opposite reason — everything is very connected in a symmetrical way.

Evidently our description of extremes so far has been rather vague and intuitive. We are going to express it in a precise and rigorous way using the concept of initial and terminal objects.

16.3 Formal definition

We will now look at the formal definitions. This is our first example of a universal property: an initial object.

Definition 16.1 An object I in a category $\mathcal{C}$ is *initial* if for every object $X \in \mathcal{C}$ there is a unique morphism $I \longrightarrow X$.

The idea is that I has a universal property among all the objects of the category: if you compare it with any object, it "comes first". This sounds a little competitive, but the idea isn't about winning really, it's more about the object helping us out as a baseline or a sort of hook on which to hang things, as it enables us to induce morphisms to other places. Imagine you're trying to record the positions of various things and you have a tape measure. It would be really helpful to have a reference point, particularly if you can attach one end of your tape measure to it. That's something like an intuition about initial objects.

Things To Think About

T 16.4 Go back to the above examples and see if you can see formally why they do or don't have an initial object.

In this category of natural numbers 0 is initial.

$$0 \longrightarrow 1 \longrightarrow 2 \longrightarrow 3 \longrightarrow \cdots$$

It might not look like there is a morphism to every other natural number, but remember that we have not drawn composite arrows in this diagram.

Formally: in this category an arrow $a \longrightarrow b$ is the assertion $a \leq b$, so we immediately know that any arrow $a \longrightarrow b$ is the unique such.[†] (An assertion just is or isn't true, so there is either one morphism or zero morphisms.) For any natural number n we know $0 \leq n$ so there is a morphism $0 \longrightarrow n$, again unique by construction.

This is an example where we proved existence and uniqueness but uniqueness actually came first; that is we first prove there is "at most one" and then that there is "at least one". We often refer to something as being "unique by construction" when that happens, and it happens quite often with universal properties.

This category of integers has no initial object.

$$\cdots \longrightarrow -2 \longrightarrow -1 \longrightarrow 0 \longrightarrow 1 \longrightarrow 2 \longrightarrow \cdots$$

This is essentially because there is no smallest integer, but formally we need to prove that for all objects a, a is not initial: for any object a there is no morphism $a \longrightarrow a - 1$, which shows a is not initial.

For the next example (now with names for the objects) there is no morphism $a \longrightarrow b$ so a can't be initial, and there is no morphism $b \longrightarrow a$ so b can't be initial.

$$\begin{tikzcd} \overset{a}{\bullet} & \overset{b}{\bullet} \end{tikzcd}$$

[†] This wording might sound odd to you. There's an implied concept at the end, which has been omitted for brevity. So the sentence really says "any arrow $a \longrightarrow b$ is the unique such arrow (that is, unique arrow $a \longrightarrow b$)".

More generally in any category with more than one object and no non-trivial morphisms there is no initial object because there just aren't enough morphisms. To have an initial object we must *at the very least* have an object with a morphism going everywhere else.

Next here's the example that "goes round in circles". We need to be careful what we mean — if f and g are inverses we will get something different from if they're a circle that loops around and never comes back to the identity.

$$\begin{tikzcd} a \arrow[r, bend left, "f"] & b \arrow[l, bend left, "g"] \end{tikzcd}$$

If the composite $g \circ f$ is *not* the identity then a can't be initial, because we have (at least) two different morphisms $a \longrightarrow a$: the composite $g \circ f$ and the identity 1_a. In fact we may have an infinite number of morphisms $a \longrightarrow a$; for example, if f and g satisfy no further equations then we will keep getting a new morphism every time we go round the loop.

On the other hand suppose f and g are in fact inverses. Now this category has exactly four morphisms: f, g, and the two identities 1_a and 1_b. All composites will be equal to one of those, as f and g compose to the identity both ways round. Then we can see that both a and b are initial.

- a is initial: there are unique morphisms $a \xrightarrow{1_a} a$ and $a \xrightarrow{f} b$
- b is initial: there are unique morphisms $b \xrightarrow{g} a$ and $b \xrightarrow{1_b} b$

Thus we have two initial objects. However, categorically speaking they're not *really* different. This is the idea of uniqueness for universal properties.

16.4 Uniqueness

There might be more than one initial object in a category, but the category "can't tell the difference" between them, in the following precise sense.

Proposition 16.2 *If I and I' are initial objects in a category $\mathcal{C}$ then they must be uniquely isomorphic, that is, there is a unique isomorphism between them.*

With proofs in higher mathematics I always think it's good to get the idea and then be able to turn the idea into a rigorous proof, rather than try and learn the proof. Also I personally just prefer understanding the idea and making it rigorous, rather than trying to read the proof and extract the idea.

The idea of this particular proof is to use the universal property of I and of I' in turn. We use the universal property of I to get a unique morphism to I', then use the universal property of I' to get a unique morphism back again, and then show that those are inverses.

Things To Think About

T 16.5 See if you can use the above idea to construct the proof yourself.

Proof Since I is initial there is a unique morphism $I \xrightarrow{f} I'$.
But also since I' is initial there is a unique morphism $I' \xrightarrow{g} I$.

We aim to show that these are inverses.

First consider the composite gf as shown.[†] $\quad I \xrightarrow{f} I' \xrightarrow{g} I$
Since I is initial there is a unique morphism $\quad I \longrightarrow I$.

But 1_I and gf are both morphisms $I \longrightarrow I$ so we must have $gf = 1_I$. (See footnote about this notation.[‡]) Similarly $fg = 1_{I'}$, thus f is an isomorphism $I \longrightarrow I'$ and is unique by construction. □

I hope you weren't put out by that "similarly" at the end. I and I' play the exact same role as each other in this proof so the part showing that $f \circ g$ is the identity is really no different from showing that $g \circ f$ is the identity, just swapping I for I' and f for g everywhere.

We can also show the converse.

Proposition 16.3 *If I' is uniquely isomorphic to an initial object I then it is also initial.*

Things To Think About

T 16.6 See if you can show this yourself. You might be able to progress just by the formalities of writing down the definitions, but remember the idea of isomorphic objects being the same is that they have the same relationships with any object in the category.

Proof Suppose I is initial, and that I' is uniquely isomorphic via f and g as shown on the right. To show I' is initial we consider any object X and exhibit a unique morphism $I' \longrightarrow X$.

$$\begin{tikzcd}
& X & \\
I \arrow[ur, "s"] \arrow[rr, bend left, "f"] & & I' \arrow[ul, dashed, "?"'] \arrow[ll, bend left, "g"]
\end{tikzcd}$$

Now, since I is initial there is a unique morphism $I \longrightarrow X$, say s, and so we have a morphism $I' \xrightarrow{g} I \xrightarrow{s} X$.

To show it's unique the idea is that if we had another one it would produce another morphism from I as well, contradicting I being initial.

† As I mentioned before, we sometimes write composites as gf to make it shorter than $g \circ f$.
‡ This unideal notation is a 1 with a subscript I, denoting the identity on I.

Consider any morphism $I' \xrightarrow{t} X$. This produces a composite $I \xrightarrow{f} I' \xrightarrow{t} X$. But I is initial so we know there is a *unique* morphism $I \longrightarrow X$.

$$\begin{array}{lrcll} \text{Thus} & s & = & t \circ f & \text{as these are both morphisms } I \longrightarrow X \\ \text{and so} & s \circ g & = & t \circ f \circ g & \text{pre-composing by } g \\ & & = & t & \text{since } f \text{ and } g \text{ are inverses.} \end{array}$$

Thus $s \circ g$ is a unique morphism $I' \longrightarrow X$ and I' is initial as claimed. □

Note that we did not need to use the fact that I and I' are *uniquely* isomorphic, as that uniqueness follows from I being initial. So in fact we have the following result.

Proposition 16.4 *Any object isomorphic to an initial object in $\mathcal{C}$ is initial.*

This categorical uniqueness for objects with universal properties is important as it means it makes sense to characterize things by universal property. If it were possible to have substantially different objects with the same universal property then it would be ambiguous to do that.

16.5 Terminal objects

So far we've only done the formal definition of universal property of objects at the "beginning" of everything. The property of being at the "end" is not really different, because it's the same idea just with all the arrows turned around. We will talk more about this in the next chapter, but it's another situation in which turning all the arrows around gets us some things for free.

Things To Think About

T 16.7 In the examples earlier we thought about the category made from the natural numbers with an arrow $a \longrightarrow b$ whenever $a \leq b$. What if we made a category with $a \longrightarrow b$ whenever $a \geq b$ instead? What does the diagram of this category look like, and what happened to the initial object?

Here we're not expressing any different information about the natural numbers, just expressing the same information in a different way.

Here's the original category of natural numbers $\quad 0 \longrightarrow 1 \longrightarrow 2 \longrightarrow 3 \longrightarrow \cdots$

and the new one: $\quad 0 \longleftarrow 1 \longleftarrow 2 \longleftarrow 3 \longleftarrow \cdots$

Visually it is apparent that we no longer have a starting point for all the arrows, but an ending point. This is the universal property that is dual to initial objects, and is called a terminal object.

Things To Think About

T 16.8 The definition of terminal object is the same as the definition of initial object, but with all the arrows turned around. See if you can write down the definition, and see that the uniqueness results immediately follow.

For the definition we just turn round the arrows in the definition of initial object, but like for monics and epics I'll also flip the diagram so that the arrows still point right, and change the letters so that we have a T for terminal.

Definition 16.5 An object T in a category $\mathcal{C}$ is *terminal* if for every object $X \in \mathcal{C}$ there is a unique morphism $X \longrightarrow T$.

The following uniqueness result can be proved as for initial objects, but with all the arrows pointing the other way. I've stated it in a slightly different form from before, because for initial objects we built this up bit by bit but now I'm saying it all at once. This form of stating a result is dry[†] but quite common, as we often want many different equivalent ways of saying the same thing.

Proposition 16.6 *Let T be a terminal object in a category $\mathcal{C}$ and T' any object. The following are equivalent:*

1. *T' is a terminal object in $\mathcal{C}$.*
2. *T' is uniquely isomorphic to T.*
3. *T' is isomorphic to T.*

What we did for initial objects was: first we showed (1) implies (2). Then we showed (3) implies (1). But (2) implies (3) by definition and so we have a loop of implications showing that the three statements are logically equivalent.

$$\begin{tikzcd}
& 2 \arrow[dr, Rightarrow, bend left] & \\
1 \arrow[ur, Rightarrow, bend left] & & 3 \arrow[ll, Rightarrow, bend left]
\end{tikzcd}$$

Things To Think About

T 16.9 Try turning all the arrows around in the cube diagrams for the factors of 30, the factors of 42, and privilege. Observe that the initial object becomes the terminal object in each case. We'll address this formally in the next chapter.

16.6 Ways to fail

Not all categories have terminal and initial objects.

Our original category of natural numbers has an initial object but no terminal object:

$$0 \longrightarrow 1 \longrightarrow 2 \longrightarrow 3 \longrightarrow \cdots$$

The version with the arrows the other way has a terminal object but no initial object:

$$0 \longleftarrow 1 \longleftarrow 2 \longleftarrow 3 \longleftarrow \cdots$$

[†] By "dry" I mean terse and expressionless, like a dry writing style.

As I've said before, instead of just declaring yes or no to whether something is true, it's illuminating to classify the different ways in which things can fail.

Things To Think About

T 16.10 Try drawing some small pictures of categories that fail to have a terminal object or fail to have an initial object. Try to isolate one issue at a time that causes the categories to fail, like when we looked at failure to be an ordered set. Also, as in that exercise, it doesn't matter what the objects of the category are; what matters is the structure of the arrows. Remember that anything that fails to have a terminal object can be entirely turned around to give something that fails to have an initial object. In some cases when you turn the arrows around you get the same thing though; make a note of when that happens as that's interesting too.

We can think about pictures and what it means for something to have no beginning or end. Or we can think about the formal definition, which involves the phrase "there exists a unique morphism", meaning there is exactly one. As usual, there are two ways to fail to have exactly one: by having more than one, or by having fewer than one (i.e. none).

That is, some categories fail to have a terminal/initial object because they have too many morphisms, and some fail because they have too few.

Going on forever

The example of the natural numbers showed us that a category can fail to have a terminal or initial object by "going on forever" in one or other direction. In that case the problem is there is no object with enough morphisms.

This category has no terminal object because no object has enough morphisms going *to* it.

$$0 \longrightarrow 1 \longrightarrow 2 \longrightarrow 3 \longrightarrow \cdots$$

Everything only has morphisms going to it from one side, not from everywhere. This is because there is no "biggest" natural number: the thing that is trying to be a terminal object is infinity, but that isn't a natural number.

In fact we can try to add in a terminal object. That turns out to be one way of fabricating something like infinity to add in to the natural numbers.

We would need an extra object, say ω, and a morphism to it from every other object, so it might look a bit like this.†

$$\begin{tikzcd}
0 \arrow[r] \arrow[drrrr] & 1 \arrow[r] \arrow[drrr] & 2 \arrow[r] \arrow[drr] & 3 \arrow[r] \arrow[dr] & \cdots \arrow[d] \\
& & & & \omega
\end{tikzcd}$$

In this diagram I have, unusually, drawn some arrows that are composites.

† The symbol ω is the Greek letter omega, which is often used to represent infinity when it's an actual object, as opposed to ∞ which is more of a concept.

- We could omit the arrow from 0 to ω as it is this composite: $0 \longrightarrow 1 \longrightarrow \omega$
- We could omit the arrow from 1 to ω as it is this composite: $1 \longrightarrow 2 \longrightarrow \omega$

$\vdots$

The trouble is we can't keep omitting arrows forever because then we won't have any arrows, so at some point we're going to have to start drawing arrows to ω. But it's most consistent to start at the beginning, and just draw them all.

This way of adding a terminal object that's missing is quite in the spirit of higher mathematics: we find ourselves in a situation that is lacking something, and that lack is preventing us from doing something we want, so we construct a new world in which we've added that thing in. This is how we get from the natural numbers to the integers and then to the rationals, the reals, and the complex numbers. In category theory this is the concept of "adding structure freely", which means you don't change anything about the structure you already had, and you don't impose any unncessary conditions on the new structure you add in. We'll come back to this in Chapter 20.

Branching

Another way to fail is by having a branching point. In that case there might be a beginning or end but not a place where *everything* begins or ends.

This example has an initial object, a, but no terminal object. The objects b and c are both ending points but because the arrows branch out, there are "too many" ending points.

$$\begin{tikzcd} & b \\ a \arrow[ur] \arrow[dr] & \\ & c \end{tikzcd}$$

Formally the problem is that b has no arrow to it from c, so b can't be terminal. But c has no arrow to it from b, so c can't be terminal. (And a has no arrow to it from either b or c.) So this is a situation where there are too few morphisms.

This one is the *dual* — I have just flipped the directions of the arrows. Thus now a is terminal but there is no initial object.

$$\begin{tikzcd} b \arrow[dr] & \\ & a \\ c \arrow[ur] & \end{tikzcd}$$

Going round in circles

We have already seen a situation of "going round in circles" that had no initial object, provided that the arrows were not inverses.

Those circles could be bigger "circles" as shown here; as long as the cycle does not compose to the identity there will be neither an initial nor a terminal object.

$$\begin{tikzcd} a \arrow[r, bend left] & b \arrow[l, bend left] \end{tikzcd} \qquad \begin{tikzcd} & b \arrow[dr] & \\ a \arrow[ur] & & c \arrow[ll, bend left] \end{tikzcd} \qquad \begin{tikzcd} b \arrow[r] & c \arrow[d] \\ a \arrow[u] & d \arrow[l] \end{tikzcd}$$

These examples are all self-dual, that is, if we turn all the arrows around the diagram is the same overall diagram (just with labels in different places).

Parallel arrows

This example has a pair of parallel arrows, meaning there are too many morphisms for a to be initial and too many for b to be terminal. This is another self-dual example.

$$a \rightrightarrows b$$

Disconnectedness

If a category is "disconnected" then there are too few arrows for anything to be initial or terminal. A category is disconnected if it separates into disjoint parts with no morphism joining them together at any point.

This example has no non-trivial morphisms, but more than one object, so no object can be terminal or initial.

$$a \qquad b$$

This example has two disconnected parts with non-trivial morphisms in each. The objects a and x are "locally initial" in their respective regions, but there is no morphism between the parts so neither can be initial overall. Likewise for b and z, with respect to terminal objects.

$$\begin{tikzcd} a \arrow[r] & b \\ x \arrow[r] & y \arrow[r] & z \end{tikzcd}$$

Empty category

The empty category has no terminal or initial object because it has no object!

Things To Think About

T 16.11 Is it possible for an object to be terminal and initial at the same time?

Here are some trivial situations in which an object is both terminal and initial.

- If the category has only one object and one (identity) morphism then the object is both terminal and initial.
- If the category has more than one object but they're all uniquely isomorphic then every object is both terminal and initial.

The second example is a useful and important type of category that we'll come back to. For now we're going to build up to seeing that the category of groups has an object that is both terminal and initial, in a more profound way.

16.7 Examples

Most of the examples we've been looking at have been abstract ones where I've just drawn categories with no particular meaning, to demonstrate structural features. Now let's look at what terminal and initial objects are in some of the examples of categories of mathematical structures that we've seen.

Sets and posets

In the category of sets and functions we've actually already seen the terminal and initial objects.

Things To Think About

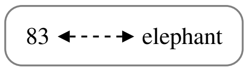
T 16.12 What are the terminal and initial objects in **Set**? Think about what set of outputs ensures that there is only one possible function *to* it, no matter what set of inputs you have (and the other way round for initial objects). A more formal but less intuitive approach is to think about the formula for the number of possible functions from one set to another, and consider what set would force that formula to give the answer 1.

You may prefer thinking the intuitive way or through the formula, but it's good to be able to do both to check them against each other. For the formula, we saw that if we have n inputs and p outputs then the number of possible functions is p^n. Now observe that $p^n = 1$ if and only if $p = 1$ or $n = 0$.

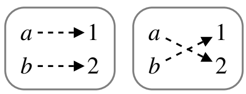
Let's think about the outputs first. Intuitively the condition $p = 1$ means we have only one output, so only one option for where any inputs go, thus only one possible function. Thus any set with only one element is terminal: if a set B has only one element, then for any set A there is exactly one function $A \longrightarrow B$.

Note that it doesn't matter what the single element is. It could be the number 1, the number 83, an elephant, or anything else. Uniqueness here is the fact that all of the one-element sets are uniquely isomorphic to one another.

There is always a unique bijection[†] between any one-element set and any other, as illustrated here.



By contrast, a pair of 2-element sets has two bijections between them, so they are isomorphic but not *uniquely* isomorphic.



If we're talking about a one-element set we often declare the single element to be an asterisk $*$ to remind ourselves that it has no meaning. The resulting one-element set is then written $\{*\}$.

Let us now think about initial objects. We saw that $p^n = 1$ if $n = 0$, and this condition means there are no inputs: the set of inputs is the empty set. We saw in Section 12.1 that this means there is always exactly one function out: it's the "empty function", which is vacuously a function as there are no inputs.

As for uniqueness of this initial object, we usually consider that there is only one empty set, as sets are defined by their elements. That is, sets A and B are

[†] Remember a bijection is a perfect matching up of objects in the two sets; formally it is injective and surjective.

usually declared equal if and only if they have the same elements, which is the case if they both have *no* elements. Thus this initial object is actually *strictly* unique. In category theory we say something is "strictly" the case to contrast it with being the case "up to isomorphism".

The terminal and initial sets are not the same. However, although they are opposite extremes categorically, they might not feel like opposite extremes as sets: they're both very small. This shows that categorical intuition may need to be developed slightly differently from other intuition.

Incidentally the terminal set is characterized by what happens when we map *into* it, but it's also a very useful set for mapping *out* of.

Things To Think About

T 16.13 What does a function $\{*\} \longrightarrow A$ consist of?

Defining a function $\{*\} \longrightarrow A$ consists of choosing one output, as we just need to know where the element $*$ is going to go. If we write 1 for the 1-element set what we're saying is: a function $1 \longrightarrow A$ "is" just an element of A.

As often happens in category theory, the "is" in the above statement needs some interpretation. Informally this means "specifying a function $1 \longrightarrow A$ is logically equivalent to specifying an element of A". If you find this a little unsatisfactory your instincts are good; there are various ways to state it more rigorously but it's good categorical practice to become comfortable with thinking of it as stated above with the word "is".

> Aside on foundations: this logical equivalence is how we can translate between category theoretic and set theoretic approaches to math. The category theoretic approach takes functions as a basic concept, rather than elements of sets, but the above shows that we can use functions out of a 1-element set to get back the concept of elements of sets, while still remaining categorical.

The basic principle of the empty set and the 1-element set can guide us to looking for initial and terminal objects in categories of sets-with-structure. It is at least a place to start looking.

Things To Think About

T 16.14 For these types of set-with-structure see if an empty one is initial, and if a 1-element one is terminal: posets, monoids, groups, topological spaces.

This is true of posets and topological spaces, but is not entirely true of monoids and groups, as we'll see in the next section. The key is that the unique function from any poset to the 1-element one is definitely order-preserving, as everything is just mapped to the same place. Similarly for the unique function from the empty poset to any other poset: there is nothing to order in the empty poset, and so the order is vacuously preserved.

Likewise for spaces, the intuition is that the unique function from any space to the one-element space is continuous because everything lands on the same point, thus everything ends up close together and nothing is broken apart. The unique function from the empty space to any other is vacuously continuous because there is nothing to map, so it is vacuously all kept close together.

Monoids and groups

Monoids and groups work a bit differently from sets, posets and spaces because they can't be empty: the definition says there must be an identity element. Indeed, the smallest possible monoid or group is the one with a single element, the identity. This is called the trivial monoid/group as it is the most trivial possible example.

Things To Think About

T 16.15 Try investigating group homomorphisms to and from the trivial group, based on what we know about functions to and from the 1-element set. Mapping *to* it, we know there's only one possible function; is it a homomorphism? Mapping *from* it, there's one possible function for each element of the target group; which ones are homomorphisms?

For the part about mapping *into* the trivial group, the point is we know that the trivial group has the terminal set as its underlying set, but is it terminal as a group? The answer is yes: the unique function to it (from anywhere else) will have to map everything to the identity, and that is definitely a homomorphism. The axioms are all equations in the target group, which only contains the identity element, so all equations are true (the two sides can only equal the identity). So the trivial group is terminal in the category **Grp**.

Now for the part about mapping *out of* the trivial group. When we're doing ordinary functions out of a 1-element set we know that we can map that single element anywhere in the target set. However, for group homomorphisms we don't have a free choice: a group homomorphism must preserve the identity element, and the single element in the source group is the identity, so it has to go to the identity in the target group. Any function picking out some other element of the target group will not be a homomorphism.

This shows there is only one possible homomorphism from the trivial group to any other, that is, the trivial group is initial in **Grp**, as well as being terminal. This is quite an interesting situation so gets a name.

Definition 16.7 A *null object* in a category is one that is both terminal and initial.

Inspired by the example of sets, terminal objects are often generically written as 1, and initial objects as 0. Thus a null object arises when $0 = 1$ in a category. This slightly shocking equation alerts us to the fact that a category with a null object behaves differently in some crucial ways. Arguably the null object for groups is one of the things that makes the category of groups such a rich and interesting place in which to develop algebraic techniques, and that is in turn why so many branches of math have advanced by putting the word "algebraic" in front of them and making some sort of connection with group theory.

Inside a poset

In a poset we can generically call the ordering "less than or equal to" as the conditions on a partial ordering make it behave like that whatever it actually is. Then an initial object corresponds to the minimum element, if there is one, and a terminal object corresponds to the maximum element.

Things To Think About

T 16.16 The minimum is an element that is less than or equal to every element. How does this correspond to an initial object?

In a poset, an arrow $a \longrightarrow b$ is the assertion $a \leq b$. A minimum, say m, is defined by being less than or equal to every element, meaning there is an arrow from m to every element. For m to be initial each of those arrows must also be unique, which is true because in a poset there is always at most one arrow $a \longrightarrow b$ for any pair of objects a, b. The fact that a maximum is terminal is analogous.

Here is a poset with an initial object but no terminal object. The element at the bottom is less than or equal to all elements, so it is initial. The ones at the top are "locally maximal" but not maximal overall, so are not terminal.[†]

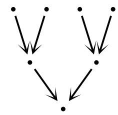
$$\begin{tikzcd}[column sep=tiny, row sep=small]
\bullet & & \bullet & & \bullet & & \bullet \\
& \bullet \arrow[ul] \arrow[ur] & & & & \bullet \arrow[ul] \arrow[ur] & \\
& & & \bullet \arrow[ull] \arrow[urr] & & &
\end{tikzcd}$$

Things To Think About

T 16.17 Turn all the arrows around in that diagram. What is the situation with terminal and initial objects now?

[†] When we say "locally" we mean something like "this is true if you zoom in to a small enough region". We could define "local maximum" formally but I hope you can see the intuition.

If we turn all the arrows around then the bottom object (physically on the page) becomes terminal and there is no longer an initial object.

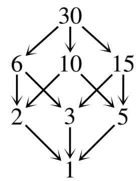


Things To Think About

T 16.18 In the diagram of generalizations of types of quadrilateral in Section 5.5 what was initial and what was terminal? What about in the diagrams for factors of numbers, and cubes of privilege?

In the diagram of quadrilaterals "square" was initial and "quadrilateral" was terminal. This was because everything was a type of quadrilateral, and squares turned out to be special cases of everything.

In the lattice/cube/poset for factors of 30 it depends which way we draw our arrows. With the directions shown on the right, 30 is initial and 1 is terminal. It might look like there is more than one arrow from 30 to 5, for example, as we could go via 10 or via 15; however, the diagram commutes and so both ways around that square are equal.



The moral of that story is that a terminal object has exactly one morphism to it from any object, but the one morphism could be expressed as a composite in more than one way.

In the corresponding diagram for privilege, rich white men are at the top, so are initial if we are drawing arrows according to *losing* one type of privilege. People with none of those three types of privilege are terminal; we previously drew the diagram with the empty set in that position, but if we put the words in, it is non-rich non-white non-men who are terminal in that particular context.

16.8 Context

One of the important things we can do with universal properties is move between different contexts and make comparisons: we can compare universal objects in different categories, or we might observe something being universal in one category but not another. We did some comparing when we looked at the initial objects in various categories of sets-with-structure.

We compared them with the initial object in **Set** and asked whether they "matched". We could think of this in terms of a schematic diagram as shown on the right.

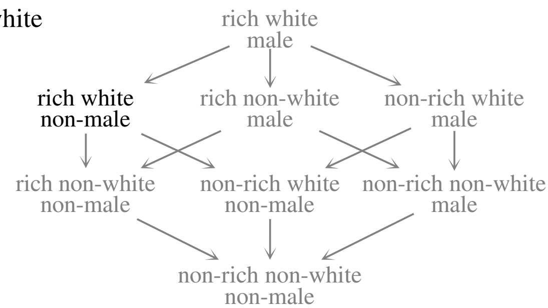
$$\begin{tikzcd}
\text{set-with-structure} \arrow[r, "\text{underlying}", "\text{set}"'] \arrow[d, dashed, "\text{initial}"'] & \text{set} \arrow[d, "\text{initial}"] \\
? \arrow[r, dashed, "\text{underlying}", "\text{set}"'] & \text{empty set}
\end{tikzcd}$$

The question then is whether we can go round the dotted arrows and still get to the empty set. For posets and topological spaces we could, but for monoids and groups the dotted arrow took us to a 1-element set, not the empty set. We will talk about this more when we talk about functors preserving structure.

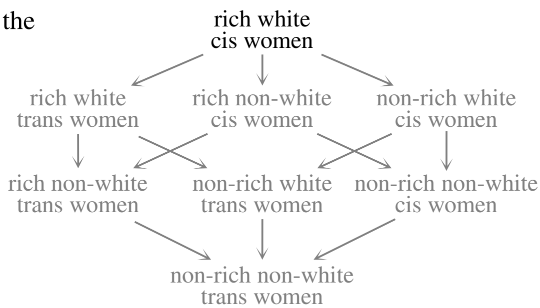
A slightly different but related question is about objects being universal (having a universal property) in one category but not another. In the case of sets compared with groups, the empty set could not become an initial object in the category **Grp** because there is no such thing as an "empty group"; a group must at least have one element, the identity. In the diagram of factors of 30, 30 is initial because we're only considering factors of 30. If we consider factors of 60 or 90, say, then 30 is there but will no longer be initial. The number 1 will still be terminal in all those categories, but if we allowed fractions then something more complicated would happen at the bottom of the diagram.

For the categories of privilege we could look in the diagram for "rich, white, male", but we could then restrict our context to women and consider the three types of privilege "rich, white, cisgender".

In the first case rich white women are not at the top.



However, in the context of just women they *are* at the top, as shown here.



Thus rich white cis women are not initial in the broader category, but they are initial if we consider a more restricted category.

I wrote about this in *The Art of Logic* as a structure that has helped me understand why there is so much anger towards white women in some parts of the feminist movement, because they only think of the first diagram and of the sense in which they are under-privileged compared with white men. This will be especially true if they tend to mix only with white people. They then forget how privileged they are relative to all other women. As with the anger of poor white men, I think it is more productive to understand the source of this anger rather than simply get angry in return.

I think it is fruitful to think about categories in which we have different universal properties so that we gain understanding of what it's like to play different roles relative to context. I can pick three types of privilege that I have: I am educated, employed and financially stable. I can also pick three types of privilege that I don't have: I am not male, I am not white, and I am not a citizen of the country I live in. I think it would be counter-productive to fixate on either of those points of view, but it is illuminating to be able to keep them both in mind (and others) and move between them.

Similarly category theory should not be about getting fixed in one category. Expressing things as categories is only the start. What's more important is moving between different categories and thereby seeing the same structure from different points of view, which is why we will soon come to functors, which are morphisms between categories.

16.9 Further topics

Mac Lane (in)famously wrote "All concepts are Kan extensions". I would say that all concepts are universal properties — the question is just *in what context*? Furthermore, all universal properties are initial objects *somewhere*. So all concepts are initial objects.

Other universal properties

There are other universal properties that we study in category theory, and we will look at some of them in Chapters 18 and 19. They can all be expressed as initial objects *somewhere*, but if a concept is very common and illuminating it can be more practical to isolate its universal property in its own right, rather than always referring to it as an initial object in some more complicated category.

Sometimes the initial objects are to do with freely generated structures, such as with monoids where the initial object, $\mathbb{N}$, was necessarily the monoid freely generated by a single element 1. Freely generated algebraic structures are related to the use of functors to move between categories. A freely generated structure is typically produced by a particular type of functor, a "free functor", which in turn has a universal property expressed via a relationship with a "forgetful functor". Those relationships are called *adjunctions*; we will not get to those in this book but I thought I'd mention the term in case you want to look them up.

Preservation by functors

I have mentioned the fact that moving between categories to change points of view is very important, and we'll come back to this in Chapters 20 and 21 on functors, the "morphisms between categories". When we do so we will ask questions about whether terminal and initial objects are preserved, that is, if we start with an initial object in one category and apply a functor, will the result still be initial in the target category of the functor?

Characterizing by universal property

In the categorical way of thinking, if a construction is somehow natural, canonical, or self-evident, it should have a universal property making that "naturalness" precise. This is a way of making our intuition rigorous. Otherwise the idea of something being natural is very flimsy because what seems natural to one person will not be at all natural to another, even among mathematicians. Exhibiting a satisfying universal property for something is satisfying in its own right, aside from it being a handy tool we can use. It can be a sort of abstract validation of our ideas.

This type of math is not about getting the right answers, and so it is not enough just to check the validity of your logic to make sure your answer is correct. This type of math is more about defining new abstract frameworks that are fruitful. If your definition is logically sound that's only the first step. If it's useful for something that's another step. But in category theory if it has a good universal property, then that shows it has internal logic of its own, and that it's not just something utilitarian that we cooked up to serve a purpose, but rather, that it grows organically from the strong roots of abstract mathematical principles.